Lemma：G是群，$H \leq G$，对$\forall a, b \in G$，有：

$\exists \lambda \in G$，s.t. $a \in \lambda H$且$b \in \lambda H \iff a^{-1}b \in H$.

Proof：($\Rightarrow$)：$\because \exists \lambda \in G$，s.t. $a \in \lambda H$且$b \in \lambda H$

$\therefore \exists h_1 \in H$，s.t. $a = \lambda h_1$

$\exists h_2 \in H$，s.t. $b = \lambda h_2$

$\therefore a^{-1}b = (\lambda h_1)^{-1}(\lambda h_2) = h_1^{-1}\lambda^{-1}\lambda h_2 = h_1^{-1}h_2 \in H$

$\therefore a^{-1}b \in H$

($\Leftarrow$)：$\because a^{-1}b \in H$　$\therefore \exists h_3 \in H$，s.t. $a^{-1}b = h_3$　$\therefore b = ah_3$

$\therefore a \in G$，且 $a = a \cdot 1_G \in aH$.

$b = ah_3 \in aH$

$\therefore a \in G$且 $a \in aH$且 $b \in aH$.　$\square$

Lemma：G是群，$A\leqslant G$，$B\leqslant G$，记 $D=A\cap B$.

设 $S_1=\{aB\mid a\in A\}$，$S_2=\{aD\mid a\in A\}$. 则有：

$$|S_1|=|S_2|=(A:D)$$

Proof：$\because G$是群，$A\leqslant G$，$B\leqslant G$ $\therefore A\cap B\leqslant G$ $\therefore D\leqslant G$

$\because G$是群，$A\cap B\leqslant G$，$A\leqslant G$，$A\cap B\subseteq A$ $\therefore A\cap B\leqslant A$

$\therefore D\leqslant A$ $\because A$是群，$D\leqslant A$ $\therefore A/D=\{aD\mid a\in A\}=S_2$

$\therefore |S_2|=|A/D|=(A:D)$

定义映射 $f: S_1\longrightarrow S_2$

$aB\longmapsto aD$

对 $\forall\, aB\in S_1$ (其中 $a\in A$). $\because a\in A$ $\therefore aD\in S_2$

$\therefore f(aB)=aD\in S_2$ $\therefore f(S_1)\subseteq S_2$

对 $\forall\, a_1B, a_2B\in S_1$ (其中 $a_1, a_2\in A$)，有：

若 $a_1B=a_2B$，则 $\because G$是群，$B\leqslant G$，$a_1, a_2\in G$ $\therefore a_1^{-1}a_2\in B$.

$\because a_1, a_2\in A$，$A\leqslant G$ $\therefore a_1^{-1}a_2\in A$ $\therefore a_1^{-1}a_2\in A\cap B$ $\therefore a_1^{-1}a_2\in D$

$\because G$是群，$D\leqslant G$，$a_1, a_2\in G$，$a_1^{-1}a_2\in D$ $\therefore a_1D=a_2D$

$\therefore f(a_1B)=a_1D=a_2D=f(a_2B)$ $\therefore f: S_1\to S_2$是一个映射.

若 $f(a_1B)=f(a_2B)$，则：$a_1D=a_2D$ $\because G$是群，$D\leqslant G$，$a_1, a_2\in G$

$\therefore a_1^{-1}a_2\in D=A\cap B$ $\therefore a_1^{-1}a_2\in B$. $\therefore a_1B=a_2B$

$\therefore f: S_1\to S_2$是一个单射.

对$\forall\, aD\in S_2$（其中$a\in A$），有：$\because a\in A$　$\therefore aB\in S_1$

$\therefore f(aB)=aD$　　$\therefore f:S_1\to S_2$是一个满射

$\therefore f:S_1\to S_2$是一个双射.　　$\therefore |S_1|=|S_2|=|A/D|=(A:D)$ $\square$

Lemma：G是群，$A\leqslant G$，$B\leqslant G$，记$S_1=\{aB\,|\,a\in A\}$，则有：

① 集合S_1的每个元素都是非空集合

② $\bigcup S_1=AB$.

Proof：对$\forall\, aB\in S_1$（其中$a\in A$）．$\because a\in G$，$B\leqslant G$　　$\therefore aB\subseteq G$.

$\because B\leqslant G$　$\therefore 1_G\in B$　　$\therefore a=a\cdot 1_G\in aB$　　$\therefore aB\neq\emptyset$.

$\therefore aB$是非空集合.　　$\therefore S_1$的每个元素都是非空集合.　①得证.

对$\forall\, x\in\bigcup S_1$，有：$\exists\,\alpha\in S_1$，s.t.　$x\in\alpha$.

$\because \alpha\in S_1$　　$\therefore \exists\, a\in A$，s.t.　$\alpha=aB$　　$\therefore x\in aB$.

$\therefore \exists\, b\in B$，s.t.　$x=ab$.　　$\because a\in A$，$b\in B$　　$\therefore x=ab\in AB$

$\therefore \bigcup S_1\subseteq AB$.

对$\forall\, x\in AB$，有：$\exists\, a\in A$，$b\in B$，s.t.　$x=ab$.

$\because a\in A$　$\therefore aB\in S_1$　$\because b\in B$　$\therefore x=ab\in aB$.

$\because aB\in S_1$且$x\in aB$　　$\therefore x\in\bigcup S_1$　　$\therefore AB\subseteq\bigcup S_1$

$\therefore \bigcup S_1=AB$.　$\square$

Remark：G是群，$A\leqslant G$，$B\leqslant G$，记$S_1=\{aB\,|\,a\in A\}$．已经证明了集合S_1的每个元素都是非空集合．已经证明了$\bigcup S_1=AB$．由选择公理知：存在函数$\psi:S_1\to AB$使得对$\forall\,\alpha\in S_1$，都有$\psi(\alpha)\in\alpha$.

Lemma: G是群, $A\leq G$, $B\leq G$, 记 $S_1=\{aB \mid a\in A\}$. 则有:

$$|S_1|\cdot|B|=|AB|.$$

Proof: 定义映射 $f: S_1\times B\longrightarrow AB$

$$(C,b)\longmapsto \psi(C)\cdot b$$

对 $\forall (C,b)\in S_1\times B$, 有: $C\in S_1$ 且 $b\in B$.

$\because C\in S_1$ $\therefore \psi(C)\in AB$ 且 $\psi(C)\in C$

$\because \psi(C)\in AB$ $\therefore \exists a_1\in A$, $b_1\in B$, s.t. $\psi(C)=a_1b_1$

$\therefore f((C,b))=\psi(C)\cdot b=(a_1b_1)b=a_1(b_1b)\in AB$

对 $\forall (C,\alpha),(D,\beta)\in S_1\times B$, 有:

若 $(C,\alpha)=(D,\beta)$, 则 $C=D\in S_1$ 且 $\alpha=\beta\in B$

$\therefore \psi(C)=\psi(D)\in AB\subseteq G$. $\therefore \psi(C)=\psi(D)\in G$.

$\because \alpha=\beta\in B\subseteq G$ $\therefore \psi(C)\cdot\alpha=\psi(D)\cdot\beta$

$\therefore f((C,\alpha))=\psi(C)\cdot\alpha=\psi(D)\cdot\beta=f((D,\beta))$

$\therefore f: S_1\times B\to AB$ 是一个映射.

若 $f((C,\alpha))=f((D,\beta))$, 则 $\psi(C)\cdot\alpha=\psi(D)\cdot\beta$

$\because \alpha\in B$, $\beta\in B$, $\psi(C)\in C$, $\psi(D)\in D$, $\psi(C)\in G$, $\psi(D)\in G$

$\therefore \psi(C)\cdot\alpha\in\psi(C)B$, $\psi(D)\cdot\beta\in\psi(D)B$

$\because C \in S_1 \; \cancel{\psi(C)B} \quad \therefore \exists\, t_1 \in A,\ s.t.\ C = t_1 B$

$\because C = t_1 B,\ t_1 \in G,\quad \psi(C) \in C,\quad \therefore C = \psi(C) B$

$\because D \in S_1 \quad \therefore \exists\, t_2 \in A,\ s.t.\ D = t_2 B$

$\because D = t_2 B,\ t_2 \in G,\ \psi(D) \in D \quad \therefore D = \psi(D) B$

$\because \psi(C)\alpha \in \psi(C) B = C,\quad \psi(C)\alpha = \psi(D)\beta \in \psi(D) B = D$

$\therefore \psi(C)\alpha \in C \cap D \quad \therefore C \cap D \neq \emptyset \quad \therefore C = D$

$\therefore \psi(C) = \psi(D) \in G \quad \because \psi(C)\alpha = \psi(D)\beta \quad \therefore \alpha = \beta$

$\therefore \cancel{(C,\alpha)=}\ (C,\alpha) = (D,\beta)$

$\therefore f: S_1 \times B \to AB$ 是一个单射.

对 $\forall \lambda \in AB$，有：$\lambda \in G$，且 $\exists\, \xi \in A,\ \eta \in B,\ s.t.\ \lambda = \xi\eta$

$\because \xi \in A \quad \therefore \xi B \in S_1$ 记 $C = \xi B \quad \therefore C \in S_1$

$\because C \in S_1 \quad \therefore \psi(C) \in AB,\ \psi(C) \in C,\ \psi(C) \in G.$

$\because G$ 是群，$B \leq G$，$\psi(C) \in G,\ \xi \in G,\ \psi(C) \in \xi B$

$\therefore (\psi(C))^{-1}\xi \in B$，令 $b = (\psi(C))^{-1}\xi \quad \therefore b \in B$

$\therefore (C, b) \in S_1 \times B$，且有：

$f((C,b)) = \psi(C) \cdot b = \psi(C) \cdot (\psi(C))^{-1}\xi = \xi$

对 $\forall \lambda \in AB$，有：$\lambda \in G$，且 $\exists \xi \in A$，$\eta \in B$，s.t. $\lambda = \xi\eta$

$\because \xi \in A$ $\therefore \xi B \in S_1$. 记 $C = \xi B$ $\therefore C \in S_1$

$\therefore \lambda = \xi\eta \in \xi B = C$ $\therefore \lambda \in C$

$\because C \in S_1$ $\therefore \psi(C) \in AB$，$\psi(C) \in G$，$\psi(C) \in C$.

$\because G$ 是群，$B \leqslant G$，$\psi(C) \in G$，$\lambda \in G$，$\xi \in G$，

$\psi(C) \in \xi B$ 且 $\lambda \in \xi B$ $\therefore (\psi(C))^{-1}\lambda \in B$.

记 $b = (\psi(C))^{-1}\lambda$ $\therefore b \in B$ $\therefore (C, b) \in S_1 \times B$

$\because f((C,b)) = \psi(C) \cdot b = \psi(C) \cdot (\psi(C))^{-1}\lambda = 1_G \cdot \lambda = \lambda$

$\therefore f: S_1 \times B \to AB$ 是一个满射.

$\therefore f: S_1 \times B \to AB$ 是一个双射. $\therefore |S_1 \times B| = |AB|$

$\therefore |AB| = |S_1 \times B| = |S_1| \cdot |B|$ $\therefore |S_1| \cdot |B| = |AB|$ $\square$

推论：G 是群，$A \leqslant G$，$B \leqslant G$，记 $D = A \cap B$，记 $S_1 = \{aB \mid a \in A\}$.

则已经证明了 $|S_1| = (A:D)$，已经证明了 $|AB| = |S_1| \cdot |B|$

$\therefore |AB| = |S_1| \cdot |B| = |B| \cdot |S_1| = |B| \cdot (A:D)$

(Remark: A 和 B 是任意两个集合，则很容易证明 $|A \times B| = |B \times A|$

$\therefore |A| \cdot |B| = |B| \cdot |A|$. 换言之，集合基数的乘法满足交换律)

Lemma: A, B, C是任意三个集合. 若 $|A| = |B|$, 则

$$|A| \cdot |C| = |B| \cdot |C|$$

Proof: $\because |A| = |B|$ $\therefore$ 存在双射 $f: A \to B$.

定义映射 $F: A \times C \longrightarrow B \times C$

$$(x, y) \mapsto (f(x), y)$$

容易验证 F是双射. $\therefore |A| \cdot |C| = |B| \cdot |C|$ $\square$

Lemma: A, B, C是任意三个集合. 则有: $(|A| \cdot |B|) \cdot |C| = |A| \cdot (|B| \cdot |C|)$

Proof: 定义映射 $f: (A \times B) \times C \longrightarrow A \times (B \times C)$

$$((a, b), c) \mapsto (a, (b, c))$$

容易验证 f是双射. $\therefore (|A| \cdot |B|) \cdot |C| = |A| \cdot (|B| \cdot |C|)$ $\square$

~~Lemma: G是群, A≤G, B≤G, 记 D=A∩B, 则有~~

Lemma: G是群, $A \leq G$, $B \leq G$, 则有:

$$|AB| \cdot |A \cap B| = |A| \cdot |B|$$

Proof: 记 $D = A \cap B$. $\because G$是群, $A \leq G$, $B \leq G$, $D = A \cap B$

$\therefore |AB| = |B| \cdot (A : D)$ $\because A \leq G$ 且 $B \leq G$ $\therefore A \cap B \leq G$

$\therefore D \leq G$. $\because G$是群, $D \leq G$, $A \leq G$, $D \subseteq A$ $\therefore D \leq A$

$\because (A,\cdot)$是群，$D \leq A$ $\quad \therefore |D|\cdot(A:D) = |A|$

$\therefore (A:D)\cdot|D| = |A| \quad \because |AB| = |B|\cdot(A:D)$

$$\therefore |AB|\cdot|D| = \big(|B|\cdot(A:D)\big)\cdot|D| = |B|\cdot\big((A:D)\cdot|D|\big) = |B|\cdot|A|$$
$$= |A|\cdot|B|$$

$\therefore |AB|\cdot|A\cap B| = |A|\cdot|B| \qquad \square$